① Let $x = 2^3 + 2^{-19} + 2^{-22}$ find machine number on the marc-32 that right on left x. Marc number

$x = (1, a_1 \dots a_{23}) * 2^m$ where $-126 \leq m \leq 127$

$x = q \times 2^m \quad 1 \leq q < 2 \quad -126 \leq m \leq 127$

$(1.a_1 \dots a_{23})_2 \times 2^m$

$x\ \text{left} = (1. \dots a_{23})_2 \times 2^m$

$x\ \text{right} = \left((1. \dots a_{23})_2 + 2^{-23}\right) \times 2^m$

$x_L \leq x \leq x\text{right}$

Answer

$$
\begin{array}{rl}
 & 0.0000000000000000001 \\
+ & 0.0000000000000000000001 \\
\hline
x = & 1{,}000.0000000000000000001001
\end{array}
$$

$1{,}0000000000000000000001001 \times 10^3$

$m = 3$

$x\text{left} = (1.0000000000100) \times 2^3$

$x\text{right} = \left((\dots\dots) + 2^{-23}\right) \times 2^3$

SORU 2

find way to compute the following function without series loss of significate.

$\log(x) - \log(y)$ Assume x and y are positive number and x is close y.

$\log(x/y) = \log(x) - \log(y)$

$\log(x/y) = \log x (x) - \log(y)$

$\log(x/((x+y)/2)) + \log(((x+y)/2)/y)$.

x close y

$\log(x/y) = \log(x/y) - \log(y/x+y)$

$\log(x/y) - \log(y/(x+y))$

SORU 3

lex $x = 0.64036768 95$

$y = 0.64036612 04$

six digit mantissa, relative error.

$x - y = 0.0000015661 \quad) a$

$0.000016 \quad) b$

Absolute $= |a-b|$

relative error $=$ absolute$/a \ast 100 = \%2.161$

4) $f(x) = x^3 - 5x^2 - x + 5$ Bisection method $a_0 = 2$ and $b_0 = 7$ calculate?

(2) − (7) +

$f(2) = 8 - 20 - 2 + 5 = -9$ (−)

$f(7) = 343 - 245 - 7 + 5 = 96$ (+)

$\boxed{f(2).f(7) < 0}$ — çıkıyorsa kök var.

4,5

adım 2

$\frac{7+2}{2} = 4,5$ $f(4,5) = 91.125 - 101.25 - 4.5 + 5 = -9.625$

adım 3

$\frac{4,5+7}{2} = 5.75$

− +

4,5 7

$f(5,75) \Rightarrow 180.109 - 165.312 - 5.75 + 5 = 24.047$

4,5 (−) 5,75 (+) (7,5)

adım 4

$\frac{4,5+5,75}{2} = 5.125$

$f(5,125) = 134.61 - 131.328 - 5.125 + 5 = 3.15$

adım 5

$\frac{4,5+5,12}{2} = 4,8$ −

$101.59 - 115.2 - 4,8 + 5 = -13.4$

adım 6

$$\frac{5,12 + 4,8}{2} = 5 \qquad 125 - 125 - 5 = \varnothing$$

kök

SORU 5

newton rapson metodu

$$x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f(x) = x^3 - 5x^2 - x + 5$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f'(x) = 3x^2 - 10x - 1$$

$$\boxed{x_0 = 7}$$

2.adım

$n = \varnothing$ için $x_1 = x_0 - \dfrac{f(x_0)}{f'(x_0)}$

$$\Rightarrow x_1 = 7 - \frac{f(7)}{f'(7)} \Rightarrow \frac{343 - 245 - 7 + 5}{147 - 70 - 1} = \frac{96}{76}$$

$$\boxed{x_1 = 5.73}$$

Galısma Notları

→ single, 8bit 23bit

Nearly machine Number

$x = q * 2^m \quad 1 \leq q < 2 \quad -126 \leq m \leq 127$

what is the number machine

$x = (1, a_1, a_2, a_3 \cdots a_{23})_2 * 2^m$

$x_+ = ((1, a_1 \cdots a_{23})_2 + 2^{-23}) * 2^m$

absolute error
real = x - y

$\text{relative} = \dfrac{\text{real} - fl(x - fl(y)}{\text{real}}$

<u>Taylor</u>

$f(x) = e^x$ in $x = 1$ taylor serisi

$f(1) = e \qquad f'(x) = e^x$

$f'(1) \cdot (x-1) = e \cdot (x-1)$

$f''(1) \cdot \dfrac{(x-1)^2}{2!} = \dfrac{e \cdot (x-1)}{2!}$

Bisection

$\dfrac{a+b}{2} = \cancel{c}$ yerine koy

$f(a) f(b) < 0$

$f(a) f(b) =$ yerine koy

Newton method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad n \geq 0$$

$$x = \sqrt{R} \Rightarrow x^2 - R = 0$$

$$f(x) = x^2 - R$$

$$f'(x) = 2x$$

gercek kök

$$|p - x_n| < TOL$$

gercek kök yoksa

$$|x_{n+1} - x_n| < TOL$$

$$\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < TOL$$

Örnek

Example

52.21875 ⇒ 110100.00111

.11010000111 × 2^6 m

23bit mattissa

0. 11010000111000000000000

sing — mattissa

127 + m⇒6 ⇒ Exponent ⇒ 133

52.21875 ⇒ 0. 11010000111 × 2^{133} → 10000101

32bit Sing used 0. 10000101

isaretsiz pozitif — sig — exp — mattissa

Example

52.21875 ⇒ 110100.00111 ⇒

1.1010000111 × 2^5 ⇒

Exponent = (e−126) = 5   e = 132